Cours de Finance

Séance 14– 19/ 02 / 2026

Financial Structure

ALBERTSCHOOL

Introduction

Semester 1: Introduction to Financial Markets

1.1 Presentation of the financial markets (4h)

1.2 Mechanisms of financial markets (2h)

1.3 Risk in finance (2h)

1.4 Equities (2h)

1.5 Bonds (2h)

1.6 Options (2h)

1.7 Role of datas in the financial markets (2h)

1.7 Financial market regulation (2h)

1.8 Case study (4h)

Semester 2 : Value

2.1 Reminder of accounting principles (2h)

2.2 Financial Analysis (2h)

2.3 Measure of creation value (2h)

2.4 Cost of capital (2h)

2.5 Financial structure (2h)

2.6 Methods of valuing a business (8h)

2.7 Case Study (4h)

2.8 Intervention (2h)

Understanding the value of a company in regards to its risk

Press Review

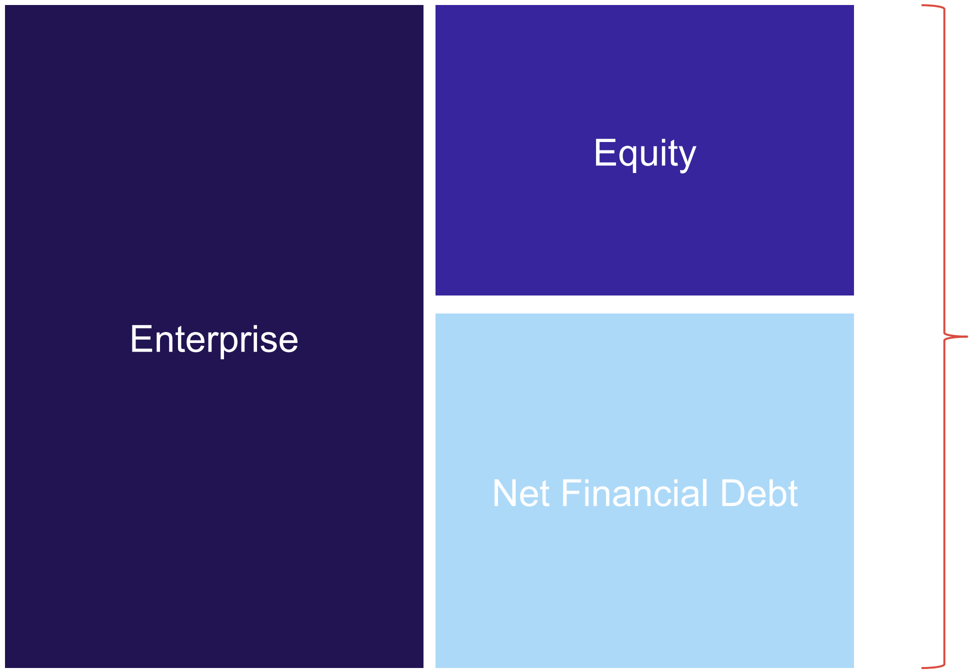
Educational content

Educational structure on financing structure :

https://www.youtube.com/watch?v=W43T00_E6H0

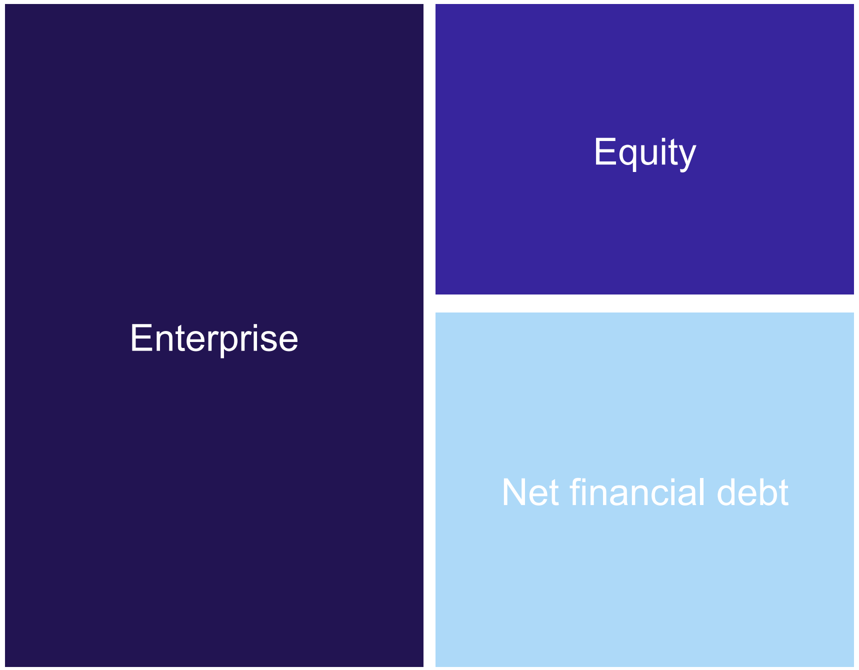
Financing structure of a company

Financing structure of a company



Capital structure refers to the breakdown between equity and debt used to finance a company's assets.

Is there an optimal capital structure?



Is there an optimal structure to maximize the value of the firm?

Is it possible to optimize WACC ?

Reminder regarding WACC formula :

$$k = kcp \times \frac{Equity}{Equity + debt} + kd \times (1 - CT\ rate) \times \frac{Debt}{Equity + debt}$$

Seeking to optimize capital structure means minimizing the cost of capital (the overall cost of financing).

Is it possible to optimize WACC ?

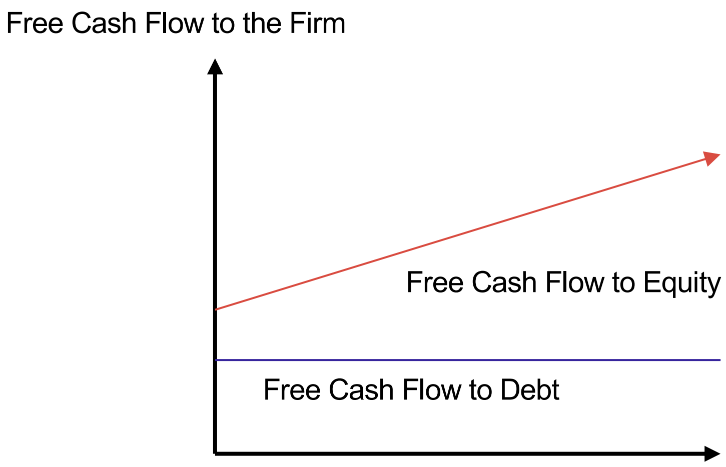
Reminder about sources of financing:

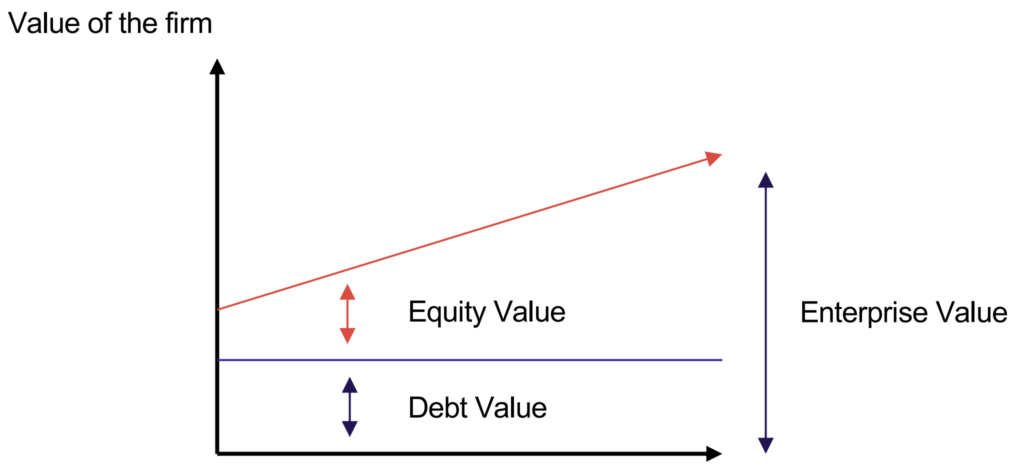
Equity: This corresponds to the funds contributed by shareholders. Its return depends on the company's profits. Shareholders have no guarantees over assets in case of default. They are the last to be repaid in the event of liquidation.

Debt: Bank or private debt has a return that is independent of performance, with repayment scheduled contractually in advance. Creditors therefore have no direct incentive linked to results. In case of default, they have guarantees allowing them to seize assets, sell them, and recover the amounts lent. They are also senior to shareholders.

⇒ We therefore understand that the risk borne by shareholders is significantly higher than that borne by creditors, so they require a higher return.

Is it possible to optimize WACC ?





The higher the value of operating assets, the more the shareholders' portion increases, while that of creditors remains unchanged.

Debt leverage

Financial leverage refers to the use of debt to increase the shareholder’s return on investment.

Example 1 :

Company 1

Enterprise 100 | Equity :100

+10% increase in operating assets

Enterprise 110 | Equity : 110

Rendement de + 10 % pour l’actionnaire

Company 2

Enterprise 100 | Equity : 10 | Net Financial Debt : 90

+10% increase in operating assets

Enterprise 110 | Equity : 20 | Net Financial Debt : 90

Rendement de + 100 % pour l’actionnaire !

Is there an optimal capital structure?

This reasoning might suggest that by maximizing leverage, we can maximize value for shareholders.

However, it should be noted that as leverage increases, the firm's risk increases as well, and therefore the cost of equity rises.

Increase in debt → Increase in cost of equity → Increase in cost of debt as well because it becomes riskier.

In 1958, Modigliani and Miller demonstrated that there is no optimal capital structure: the cost of capital remains constant regardless of the firm's leverage level.

In a way, the value of an apartment does not depend on how it is financed.

Example

	Company X	Company Y
EBIT	20 000	20 000
Interest on debt (5%)	0	4 000
Net income	20 000	16 000
Dividends = net income	20 000	16 000
Cost of Equity (Ke)	10%	12%
Equity Value (Div / Ke)	200 000	133 333
Debt Value (Interest / Cost of Debt)	0	80 000
Enterprise Value	200 000	200 000
WACC	10%	9,4%
Leverage	0%	60%

Is there an optimal capital structure?

At first glance, it might seem preferable to invest in company Y, which has higher leverage.

Modigliani and Miller showed that a shareholder of company Y could replicate and optimize the same investment by buying shares of company X.

How is this possible?

Is there an optimal capital structure?

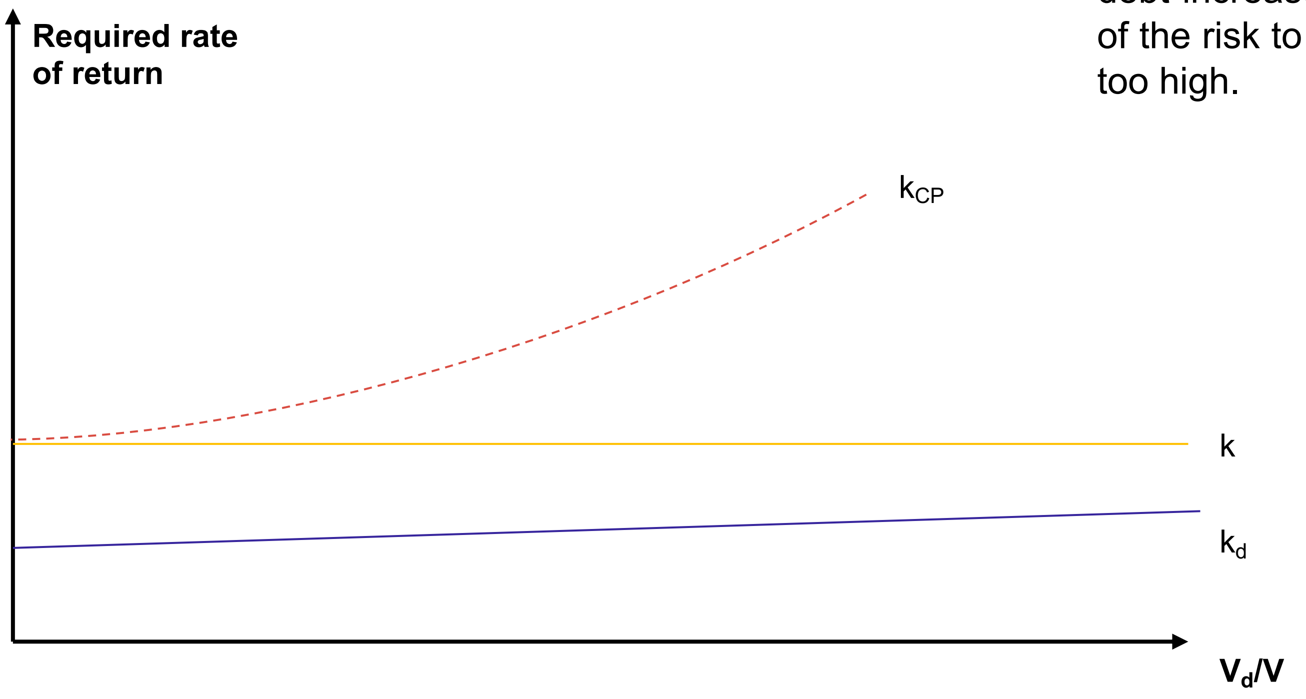
Assume a shareholder of company Y owns 1% of the company, i.e. 1% × 133,333 = 1,333.

1/ He sells his shares for 1,333
2/ He borrows personally at 60% (the same leverage as company Y). He therefore obtains financing of 60% × 1,333 = 800 at 5% interest
3/ His investment capacity becomes 1,333 + 800 = 2,133
4/ He buys shares of company X
5/ His income was 12% × 1,333 = 160, and now it is 10% × 2,133 − 800 × 5% = 173

The value of operating assets is therefore independent of how they are financed.

Is there an optimal capital structure?

The pace of the cost of equity slows as debt increases. Shareholders transfer part of the risk to creditors when debt becomes too high.



Is there an optimal capital structure?

A company's capital structure results from many factors unrelated to firm value, such as taxation or conflicts of interest between shareholders and managers.

Financing structure and tax

Unlike dividends, interest expenses are tax-deductible. Modigliani and Miller revisited their demonstration by including the tax shield, showing that it becomes more advantageous to favor debt.

Example :

Company 1 :

Required rate of return for shareholders: 10% → 10.
For a 25% tax rate, the company must generate a pre-tax profit of 13.33.

Company 2 :

Required rate of return for shareholders: 14% → 5.6.
Interest rate: 5%
For a 25% tax rate, the company must generate a pre-tax and interest profit of 10.47.

Financing structure and tax

However, several remarks must be considered:

⇒ This assumes the firm continuously generates profits and therefore pays corporate tax. If it is loss-making, there is no additional tax benefit.
⇒ The firm must generate sufficient earnings to cover financial expenses.
⇒ The risk of bankruptcy linked to excessive leverage must also be taken into account.

Value of an unlevered firm
+ Present value of tax shields from debt
− Present value of bankruptcy and distress costs
= Value of a levered firm

Debt as a means to pressurize managers

A manager who is not a shareholder has every incentive to avoid debt as much as possible, since it adds risk to the firm he manages. He may even prefer to accumulate cash and not invest it in growth.

This can sometimes lead to situations where excess cash is used for managerial perks (company cars, luxurious offices, etc.).

Using debt therefore allows shareholders to discipline managers by imposing repayment obligations and encouraging them to maximize returns under the constraint of leverage.